

NeuralBench: A Unifying Framework to Benchmark NeuroAI Models

Hubert Banville¹, Stéphane d’Ascoli¹, Simon Dahan¹, Jérémy Rapin¹, Marlène Careil¹, Yohann Benchetrit¹, Jarod Lévy^{1,4}, Saarang Panchavati¹, Antoine Ratouchniak¹, Mingfang (Lucy) Zhang^{2,3}, Elisa Cascardi¹, Katelyn Begany¹, Teon Brooks¹, Jean-Rémi King¹

¹Brain & AI team, Meta FAIR, ²École Normale Supérieure, Université PSL, CNRS, ³Hospital Foundation Adolphe de Rothschild, ⁴MIND, Inria

Deep learning and large public datasets have recently catalyzed the proliferation of AI models for processing brain recordings. However, systematically evaluating these models remains a challenge: not only do the preprocessing pipelines, training and finetuning approaches largely vary across studies, but their downstream evaluation is often limited to small sets of tasks and/or datasets. Here, we present NeuralBench: a unified framework for benchmarking AI models of brain activity. We accompany this framework with NeuralBench-EEG v1.0 – a large EEG benchmark that includes 36 electroencephalography (EEG) tasks and 14 deep learning architectures, and is evaluated on 94 datasets accessed through a standardized interface. This first EEG-focused release already highlights two main findings. First, current foundation models only marginally outperform task-specific models. Second, a large set of tasks (*e.g.* cognitive decoding, clinical predictions) remain highly challenging, even for the best models. Critically, NeuralBench is designed for the integration of new tasks, datasets, models, and neuroimaging modalities, as illustrated by preliminary extensions to MEG and fMRI datasets and models. Through this white paper, we invite the community to expand this open-source framework and work together toward a unified benchmarking standard for neuroimaging models.

Correspondence: hubertjb@meta.com, jeanremi@meta.com

Code: <https://github.com/facebookresearch/neuroai/tree/main/neuralbench-repo>

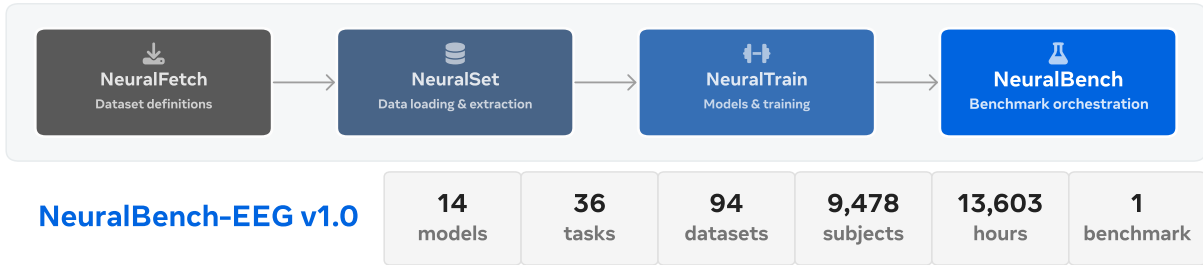


Figure 1 NeuralBench is a benchmarking framework for the evaluation of brain models, based on the NeuralFetch, NeuralSet and NeuralTrain libraries, and providing a fully configurable and flexible interface. **Second row.** Contents of the first release, NeuralBench-EEG v1.0.

1 Introduction

The rise of AI models in neuroimaging. Historically focused on targeted, small-scale, and private studies (Poldrack, 2011; Michel and Murray, 2012; Baillet, 2017), the analysis of brain activity is now rapidly shifting toward large-scale public datasets (Van Essen et al., 2013; Harati et al., 2014; Taylor et al., 2017; Pinho et al., 2018; Sun et al., 2025; d’Ascoli et al., 2026). On top of boosting statistical power and stimulating reproducible research (Button et al., 2013; Poldrack et al., 2017; d’Ascoli et al., 2026), the proliferation of open datasets has catalyzed the use of AI to process brain recordings. Specifically, the self-supervised learning methods

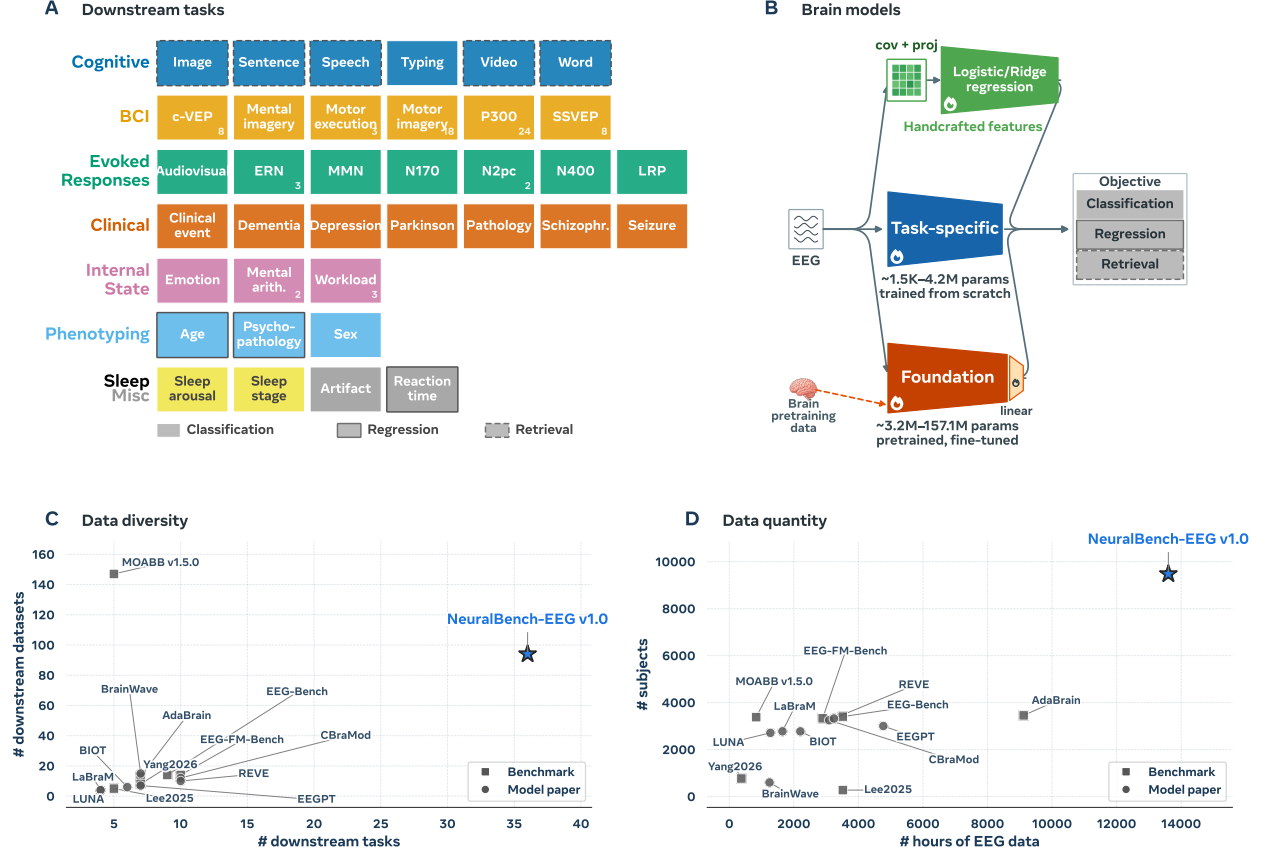


Figure 2 Detailed overview. **A.** NeuralBench-EEG v1.0 currently includes 36 decoding downstream tasks on 94 different datasets, covering 8 different categories such as cognitive decoding, brain-computer interfacing (BCI), evoked responses and clinical tasks. **B.** Three types of models are compared: task-specific from-scratch deep learning architectures, pretrained foundation models and handcrafted features-based baselines. **C-D.** NeuralBench-EEG v1.0 currently contains more than three times the number of tasks of previous EEG downstream task benchmarks, and more than twice as many individual subjects, making it the largest such open-source benchmark yet.

developed for language, speech and images (Devlin et al., 2019; Baevski et al., 2020; Oquab et al., 2024) are now adapted for the development of *brain foundation models* (Yang et al., 2025; Zhou et al., 2025). These models start to transfer to downstream tasks spanning electroencephalography (EEG) (Jiang et al., 2024; Kostas et al., 2021; Yang et al., 2023; Wang et al., 2024; Döner et al., 2025; El Ouahidi et al., 2025), functional magnetic resonance imaging (fMRI) (Kim et al., 2023; Dong et al., 2024; Caro et al., 2023), intracranial EEG (Wang et al., 2023; Yuan et al., 2024), structural MRI (Tak et al., 2026), electrophysiology (Azabou et al., 2023), and calcium imaging (Wang et al., 2025a).

Current benchmarking challenges. Despite these advances, the evaluation landscape is fragmented. For EEG, the Mother of All BCI Benchmarks (MOABB)¹ (Jayaram and Barachant, 2018; Aristimunha et al., 2025) provides an extensive benchmark for brain-computer interfacing (148 datasets). However, it focuses on a narrow set of 5 downstream tasks (e.g., left versus right motor imagery) and primarily targets handcrafted features-based approaches, such as Riemannian geometry pipelines for trial classification (Barachant et al., 2011; Yger et al., 2016). Building on this, Borra et al. (2024) adapted the benchmark for deep learning, focusing specifically on a subset of nine BCI dataset from the MOABB and three well-established architectures. In parallel, benchmarks tailored to EEG *foundation* models have emerged, e.g., EEG-Bench (Kastrati et al., 2025), EEG-FM-Bench (Xiong et al., 2025), and AdaBrain-Bench (Wu et al., 2025). However, these efforts too are limited to a restricted set of downstream evaluations (Figure 2). This fragmentation is even more

¹<https://moabb.neurotechx.com/>

pronounced in other brain recording modalities. For intracranial EEG (iEEG), Karpowicz et al. (2024) introduced FALCON, a few-shot evaluation benchmark for invasive BCI tasks, but is limited to 5 evaluation datasets. Elsewhere, foundation models for fMRI, MEG, and other modalities are still evaluated on handpicked datasets without a unified, systematic benchmark. Overall, there is currently no comprehensive framework for standardizing and efficiently evaluating brain foundation models across neuroimaging modalities.

Contribution. Here, we introduce NeuralBench, a community-driven benchmarking framework for evaluating foundation models on brain activity recordings. NeuralBench is designed to be:

1. **Comprehensive.** As many relevant downstream tasks as possible should be included, for as many relevant brain imaging devices as possible. This helps properly support claims of models being truly foundational and provides a more nuanced assessment of the strengths and weaknesses of different models.
2. **Rigorously validated.** Publicly available datasets often require careful curation and tailored onboarding efforts so their contents can be used properly. Similarly, the framing of learning tasks requires special care to avoid common pitfalls (*e.g.* leakage, overfitting, etc.) and to be maximally informative. This also applies to foundation models – each one is trained using a specific preprocessing pipeline that needs to be replicated to allow fair evaluation.
3. **Standardized.** By providing a unified point-of-entry for diverse brain imaging devices (EEG, iEEG, MEG, fMRI, fNIRS, etc.), NeuralBench makes it easy to build and evaluate multimodal pipelines, bringing us closer to integrated brain modeling.
4. **Flexible.** NeuralBench is configurable at all levels of the neuroimaging modeling pipeline (data source, preprocessing, architecture, optimization, metrics, etc.), offering a compact and flexible interface for running and evaluating brain models.

The first release of NeuralBench, **NeuralBench-EEG v1.0**, supports 36 tasks evaluated on 94 EEG datasets (with some tasks spanning multiple datasets), accessed through a standardized interface. This effort, which capitalizes on the careful curation of public datasets, provides unique support for cognitive decoding tasks (*i.e.*, image, sentence, speech, typing, video and word decoding), which, apart from image decoding, have been omitted in previous benchmarks. In addition, we demonstrate the versatility of this framework and seed its expansion towards other brain imaging modalities by including a couple of illustrative MEG and fMRI tasks and models.

Goal and overview. In the following sections, we present a technical overview of NeuralBench, followed by initial results on **NeuralBench-EEG v1.0** which highlight two main findings: current EEG foundation models only marginally outperform task-specific baselines, and many complex tasks (*e.g.*, cognitive decoding, clinical prediction) remain challenging even for foundation models. We then show how NeuralBench seamlessly extends to MEG and fMRI modalities.

Toward a Community-Driven Standard. Recognizing that a unified standard is only as strong as the community behind it, we provide the NeuralBench codebase² as a starting point for global collaboration. We invite researchers across tasks and modalities to co-author its evolution and join us in defining the future of brain model evaluation.

2 Methods

In this section, we give a technical overview of NeuralBench, including its general architecture, a description of supported downstream tasks, model architectures, adaptation strategies, and evaluation protocols. We provide a high-level summary here and refer the reader to the package documentation for additional technical details.

²<https://github.com/facebookresearch/neuroai/tree/main/neuralbench-repo>

2.1 Benchmark architecture

NeuralBench is built on top of three main Python packages, which provide an interface with the data, processing and modeling tools that already exist in the neuroscience software ecosystem. Specifically, public neuroimaging datasets are accessed through NeuralFetch, which fetches curated datasets from public repositories like OpenNeuro³, DANDI⁴, and NEMAR⁵. Then, neural data is prepared and made available as PyTorch-ready dataloaders with NeuralSet– a package designed to efficiently leverage existing packages like MNE-Python (Gramfort et al., 2013)⁶ and nilearn (Abraham et al., 2014)⁷ for applying classic neuroimaging preprocessing pipelines, as well as HuggingFace⁸ for extracting AI model embeddings (*e.g.* on word, speech and image modalities). Finally, NeuralTrain provides modular training code with PyTorch-Lightning⁹, Pydantic¹⁰ and exca (Rapin and King, 2024)¹¹ (see Figure 2D).

Once installed, NeuralBench is run with a command-line interface (CLI). For example, the following can be used to run the **audiovisual stimulus** classification task on the MNE-Python sample dataset (Gramfort et al., 2013):

1. Install NeuralBench

```
$ pip install neuralbench
$ neuralbench --help # Trigger
                        interactive path configuration
```

2. Run the audiovisual stimulus task on EEG

```
$ neuralbench eeg audiovisual_stimulus --download # Download data
$ neuralbench eeg audiovisual_stimulus --prepare  # Prepare cache
$ neuralbench eeg audiovisual_stimulus           # Run the task
```

On top of allowing quick sanity-checking of the codebase, this small dataset (<300 examples) is an interesting case study for very-low-data regime downstream tasks.

Task configuration Tasks are configured through lightweight YAML files, which describe data sources, train/validation/test splits, neural data preprocessing (filtering, scaling, windowing, etc.), target processing if any (*e.g.* extracting dense representations of rich stimuli), trainer configuration (number of epochs, learning rate, etc.), and metrics.

Example YAML configuration for audiovisual stimulus classification on EEG.

```
data:
  study:
    source.name: Mne2013SampleEeg
  split:
    name: SklearnSplit
    valid_split_ratio: 0.2
    test_split_ratio: 0.2
    stratify_by: description
  neuro.baseline: [0.0, 0.2]
  target:
    name: LabelEncoder
    event_types: Stimulus
    event_field: description
    return_one_hot: true
  trigger_event_type: Stimulus
  start: -0.2
  duration: 1.0
  loss.name: CrossEntropyLoss
  metrics: BalancedAcc
```

³<https://openneuro.org/>

⁴<https://dandiarchive.org/>

⁵<https://nemar.org/>

⁶<https://mne.tools/>

⁷<https://nilearn.github.io/>

⁸<https://huggingface.co/>

⁹<https://lightning.ai>

¹⁰<https://pydantic.dev/>

¹¹<https://github.com/facebookresearch/exca>

Model configuration Similarly, deep learning architectures are defined through YAML files that configure NeuralTrain models. Thanks to its configurable pydantic layer, NeuralTrain can transparently access models from other libraries such as BrainDecode¹² (Schirrneister et al., 2017; Aristimunha et al., 2026).

2.2 Task and dataset selection

A primary objective of NeuralBench is to provide a wide coverage of downstream tasks. Focusing on EEG for this first release, we selected a set of 36 representative downstream tasks, covering 8 categories: cognitive decoding, brain-computer interfacing (BCI), evoked responses, clinical, internal state, sleep, phenotyping and miscellaneous (see Figure 2A). Note that each task does not necessarily fall neatly into one of these categories, *e.g.* the P300 classification task could also fall under the *Evoked responses* category, however since it is usually collected as part of a P300 speller protocol we file it under *BCI*. Similarly, the *psychopathology* classification task could go under the *Clinical* category, but we opt to keep it under the same *Phenotyping* category as other tasks based on Shirazi et al. (2024). Nevertheless, we keep this operational organization to facilitate interpretation.

Several inclusion criteria were considered. First, whenever possible, openly accessible datasets, rather than datasets requiring additional manual steps, were prioritized. Second, as several EEG datasets are approaching saturation (*e.g.* binary *pathology detection* on the TUAB dataset (Lopez et al., 2015)), we devoted special efforts to include challenging tasks on which performance is likely to be significantly improvable. To this end, we include multiple *cognitive decoding* tasks, where the goal is to recover a dense representation of the stimulus presented to participants, *e.g.* speech, images, words and videos (Défossez et al., 2022; Benchetrit et al., 2024; d’Ascoli et al., 2025). Finally, we re-framed common EEG classification tasks, *i.e.*, *clinical event* classification on the TUEV dataset (Harati et al., 2015) and *artifact* classification on the TUAR dataset (Hamid et al., 2020), as *multilabel* classification tasks to take into account that that some events can overlap.

Note on pretraining data overlap. Existing EEG foundation models are trained on custom corpora curated by the model authors. As there is a limited set of publicly available EEG datasets, some of these models saw (part of) the downstream datasets during pretraining. We flag these instances (*e.g.* with hashed bars in Figure 4), rather than discard them, as we did not notice a clear trend suggesting pretraining “leakage” improves downstream performance on the same data.

Finally, while we prioritized breadth over depth in our selection of datasets (*i.e.*, most tasks rely on a single core dataset), we also leveraged readily available datasets through MOABB¹³ (Aristimunha et al., 2025) to offer multiple comparison points on a few tasks, such as *motor imagery* and P300 decoding. This gives rise to two benchmark variants: NeuralBench-EEG-Core v1.0, which focuses on a single dataset per task, and NeuralBench-EEG-Full v1.0, which has up to 24 datasets per task and therefore enables the study of within-task variability.

In total, NeuralBench-EEG v1.0 covers 36 tasks, 94 datasets, 9,478 subjects and 13,603 hours (see Table 1). Of note, no data is contained in the released package; datasets are accessed through the standardized interface provided by NeuralSet and NeuralFetch (King et al., 2026). Also, all datasets sources already provide de-identified data; as a result, the benchmark only processes de-identified subject IDs. More details on datasets and downstream tasks can be found in Appendix A and in the benchmark documentation¹⁴.

2.3 Model architecture selection

We selected commonly used EEG architectures and foundation models. First, we included 8 task-specific neural network architectures (Table S2), *i.e.*, models that are trained from scratch on downstream tasks, typically with lower parameter counts. These include ShallowFBCSPNet and Deep4Net (Schirrneister et al., 2017), EEGNet (Lawhern et al., 2018), BDTCN (Gemein et al., 2020), ATCNet (Altaheri et al., 2022), EEGConformer (Song et al., 2022), SimpleConvTimeAgg (El Ouahidi et al., 2023) and CTNet (Zhao et al., 2024).

¹²<https://braindecode.org/>

¹³<https://moabb.neurotechx.com/>

¹⁴<https://github.com/facebookresearch/neuroai/tree/main/neuralbench-repo>

Table 1 Tasks and datasets in **NeuralBench-EEG v1.0**. For each task, we indicate the *Core* dataset name, the machine learning objective (*reg*: regression, *clf*: classification, *retr*: retrieval), the number of unique EEG channels, the number of unique subjects, the dimensionality of the target (*i.e.*, the number of outputs), and the number of additional datasets in the *Full* variant, if any.

Task	Core dataset	Obj.	#Ch.	#Subj.	#Hours	#Out.	+Datasets
Age	Shirazi2024	reg	129	2,858	95.3	1	
Artifact	Hamid2020	clf	26	213	300.2	5	
Audiovisual stimulus	MneSample2013	clf	60	1	0.1	4	
Clinical event	Harati2015	clf	21	370	20.1	6	
C-VEP	Thielen2021	clf	8	30	50.0	20	7
Dementia diagnosis	Miltiadous2023	clf	19	88	19.5	3	
Depression diagnosis	Mumtaz2018	clf	19	64	20.4	2	
Emotion	Chen2023	clf	32	123	28.7	9	
Error-related negativity	Kappenman2021Ern	clf	30	40	4.4	2	2
Image	Gifford2022	retr	63	10	230.7	1,536	
Mental arithmetic	Zyma2019	clf	20	35	2.3	2	1
Mental imagery	Scherer2015	clf	30	9	3.9	5	
Mental workload	Hinss2023	clf	63	29	21.4	3	2
Mismatch negativity	Kappenman2021Mmn	clf	30	40	11.1	2	
Motor execution	Srisrisawang2024	clf	60	20	16.0	16	2
Motor imagery	Stieger2021	clf	60	62	141.5	4	17
N170	Kappenman2021N170	clf	30	40	3.6	2	
N2pc	Kappenman2021N2pc	clf	30	40	3.6	2	1
N400	Kappenman2021N400	clf	30	40	1.3	2	
P300	Schreuder2010	clf	60	21	70.5	2	23
Parkinsons diagnosis	Singh2021	clf	60	129	54.9	2	
Pathology	Lopez2017	clf	21	2,329	971.7	2	
Psychopathology	Shirazi2024	reg	129	2,820	94.0	1	
Reaction time	Shirazi2024	reg	129	1,945	70.8	1	
Lateralized readiness potential	Kappenman2021Lrp	clf	30	40	4.4	2	
Schizophrenia diagnosis	Albrecht2019	clf	62	77	34.6	2	
Seizure	Dan2023	clf	24	23	982.9	2	
Sentence	Hollenstein2018	retr	128	12	21.5	768	
Sex	Shirazi2024	clf	129	2,858	95.3	2	
Sleep arousal	Ghassemi2018	clf	6	994	7,662.3	2	
Sleep stage	Kemp2000	clf	2	78	1,635.2	5	
Speech	Brennan2019	retr	60	33	58.5	4,096	
Typing	Levy2025	clf	61	20	40.7	29	
Video	Liu2024	retr	62	20	16.3	1,408	
Steady-state visually evoked potential	Wang2017	clf	64	34	9.1	40	6
Word	Nieuwland2018	retr	74	222	441.4	1,024	

Second, we selected 6 EEG foundation models (Table S4), *i.e.*, larger models that have been pretrained on unlabelled data, and that are typically finetuned on downstream tasks. These are BENDR (Kostas et al., 2021), LaBraM (Jiang et al., 2024), BIOT (Yang et al., 2023), CBraMod (Wang et al., 2025b), LUNA (Döner et al., 2025) and REVE (El Ouahidi et al., 2025).

Third, we included representative handcrafted features-based models using sklearn-like pipelines to provide a non-deep learning comparison point (see Table S3). These pipelines all rely on symmetric positive definite (SPD) matrix representations of the input EEG which are fed to ‘shallow’ predictors (logistic or Ridge regression).

Finally, we include chance-level and “dummy” baselines to offer a proper performance floor to which models can be compared. Chance-level performance corresponds to the performance obtained by an untrained, randomly initialized model. “Dummy” performance corresponds to the performance obtained by a model that predicts either the average target for regression and retrieval tasks, the most frequent class for binary and multiclass classification tasks, or samples class-wise labels based on training set statistics for multilabel classification tasks. The dummy models are always fitted on the same training and validation sets as the other models.

2.4 Downstream strategy protocol

The original publications presenting the EEG foundation models listed above use different adaptation recipes, *i.e.*, for optimization (batch size, optimizer, learning rate, scheduler, etc.), aggregation ([CLS] token, average patch token, etc.) and projection head design (linear layer, MLP, etc.). Here, to focus the comparison on model architecture and pretraining methodology, we adopt a shared training recipe.

Each foundation model is finetuned end-to-end, using a linear projection head which receives the average-pooled output tokens and outputs a single vector matching the target dimension. Training uses AdamW (Loshchilov and Hutter, 2017) with a learning rate of 10^{-4} , weight decay of 0.05, and cosine-annealing (10% warmup), for up to 50 epochs with early stopping on the corresponding validation metric (patience=10). The sole exception to this unified recipe is BENDR (Kostas et al., 2021), for which we lower the learning rate to 10^{-5} and apply gradient clipping at 0.5 to obtain stable learning curves. Therefore, model-specific techniques from the original papers — such as layer-wise learning rate decay, two-stage probing, or LoRA (Hu et al., 2022) — are intentionally omitted and the comparison of downstream adaptation strategies is kept for future iterations of NeuralBench. Finally, task-specific architectures are trained *from scratch* using randomly initialized weights, using the same optimization recipe as the foundation models.

2.5 Preprocessing

We take particular care to follow the preprocessing pipelines described in the original EEG foundation model papers, as pretrained checkpoints expect specific input statistics which require identical preprocessing. For task-specific architectures, we use a single preprocessing configuration inspired by previous EEG decoding work (Défossez et al., 2022; Benchetrit et al., 2024): resampling to 120 Hz, bandpass filtering between 0.1 and 75 Hz, notch filters at 50 and 60 Hz as well as their harmonics, channel-wise robust scaling at the recording level, followed by clamping at 20. Nevertheless, it will be important, in future iterations, to systematically evaluate the impact of preprocessing strategies on downstream evaluation.

2.6 Data Splitting

The generalization capacity of a model can be highly dependent on the task. Consequently, to split datasets into training, validation and testing sets, we use different strategies for each task: (1) predefined splits when provided by the authors, (2) “leave-concept-out” for cognitive decoding tasks, (3) cross-subject splits, with optional stratification for clinical tasks, or (4) random splits when very few examples are available. We train (or finetune) each model on each task three times, using a single train/valid/test partition but three different random seeds for the initialization of the (non-pretrained) model weights. See Appendix C for more details.

2.7 Training and evaluation metrics

The training, finetuning and evaluation of the models must vary depending on the task objective. In binary and *multiclass* classification tasks, we use cross-entropy with optional weighing to mitigate class imbalance. In *multilabel* classification tasks, we instead use binary cross-entropy per target. Regression tasks rely on the mean squared error (MSE) loss. Finally, we also implemented retrieval tasks, where a model must recover which candidate target out of a given retrieval set actually corresponds to the input (*e.g.* which image out of a set of multiple images seen by a participant, based on an input EEG window). Retrieval is a particularly well-posed problem for cognitive tasks, as the content of an image, text or video is neither categorical, nor necessarily perfectly specified in a pretrained embedding. Retrieval ultimately allows future models and approaches to be compared with one another, with little assumption on how to represent the target. For retrieval, we here rely on the CLIP loss (Radford et al., 2021), using the configuration of Défossez et al. (2022), *i.e.*, with the brain-to-target term only and a fixed temperature parameter $\tau = 1$:

$$\mathcal{L}_{CLIP}(\theta) = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\hat{\mathbf{z}}_i, \mathbf{z}_i)/\tau)}{\sum_{j=1}^B \exp(s(\hat{\mathbf{z}}_i, \mathbf{z}_j)/\tau)} \quad (1)$$

where B is the batch size, s is the cosine similarity, and $\mathbf{z}_i$ and $\hat{\mathbf{z}}_i$ are the target dense embedding (*e.g.* DINOv2 (Oquab et al., 2024) for image retrieval) and corresponding prediction for batch element i .

2.8 Model comparison

To help summarize the comparison across models, we opted to report a single representative metric per task type: *i.e.*, balanced accuracy for binary and multiclass classification, F1-score for multilabel classification, Pearson correlation for regression, and top-5 accuracy for retrieval¹⁵. See Appendix D for more details. We further report normalized scores $\tilde{s}$ to facilitate cross-task comparisons and emphasize task difficulty:

$$\tilde{s} = \frac{s - s_{\text{dummy}}}{s_{\text{perfect}} - s_{\text{dummy}}} \quad (2)$$

where s_{dummy} is the performance obtained by the dummy model on a specific dataset, and s_{perfect} is the metric-specific theoretical highest performance obtainable (*e.g.* 100% balanced accuracy for classification tasks). Of note, due to the inherent noise in brain activity recordings, this perfect score cannot be attained in practice. With this formulation, a value of 0 indicates dummy-level performance and a value of 1 indicates perfect performance.

Finally, global model rankings consists of the per-task rank of each model, averaged over all tasks. On NeuralBench-EEG-Full v1.0, we additionally use all available datasets to compute this ranking, *i.e.*, we average ranks across datasets of the same task first.

2.9 Computational considerations

NeuralBench-EEG-Full v1.0 requires running 4,947 experiments. (14 models + chance, dummy & handcrafted features baselines, 97 tasks-datasets, 3 seeds each). By default, each job uses a single GPU with at least 32 GB VRAM and 64 GB of CPU RAM, though measured peak GPU usage averages only ~ 1.3 GB (max ~ 30.3 GB). The full EEG benchmark requires ~ 11 TB of disk space (~ 3.2 TB for the raw data, 7.8 TB of preprocessed cache, and 333 GB for logged results). Training times vary widely, with a median of 2.7 minutes but mean of 21.7 minutes per experiment, with the longest run taking 18.2 hours. This yields an estimated total serial runtime of ~ 73 days (1,751 GPU-hours). The benchmark runs on Python ≥ 3.12 with PyTorch 2.6 (Paszke et al., 2019) and Pytorch-Lightning (Falcon and The PyTorch Lightning team, 2019).

¹⁵We evaluate top-5 accuracy on the entire test set, using within-subject aggregation of repeated predictions, if available.

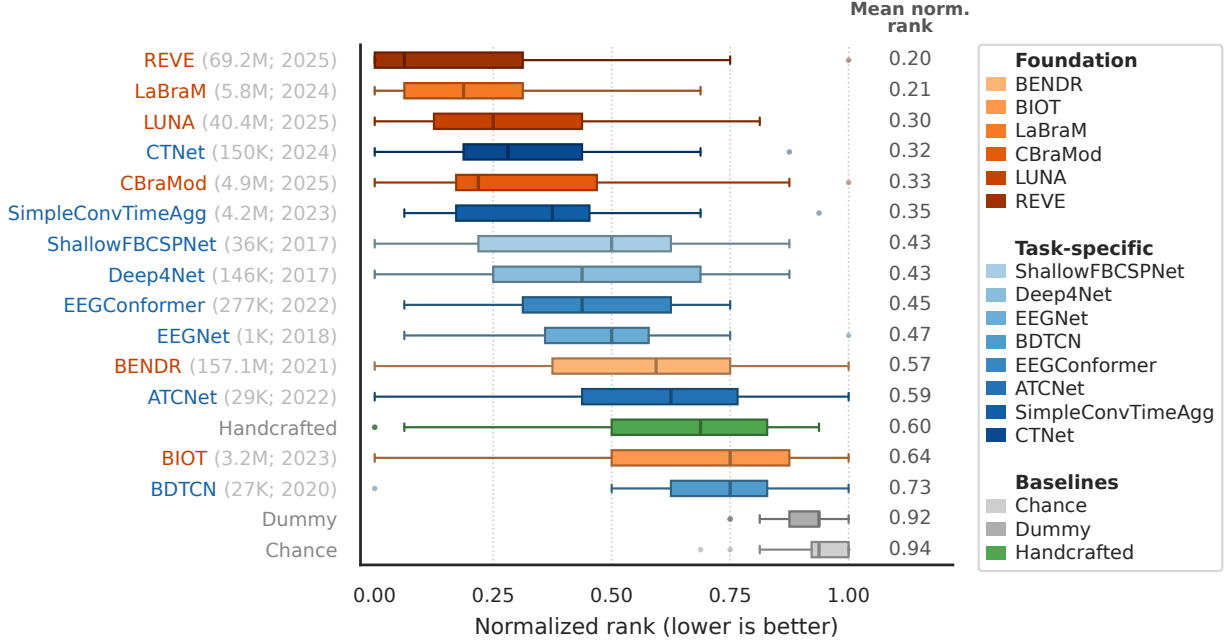


Figure 3 Model ranking on NeuralBench-EEG-Core v1.0. Each box corresponds to the rank distribution of one model across downstream tasks. Models are ordered by their mean normalized rank, displayed on the right hand side of the boxes. The number of trainable parameters (excluding the last linear projection layer) and the year of original publication are shown in parentheses next to each model name. Recent foundation models outrank most smaller task-specific models, suggesting large self-supervised pretraining can benefit downstream performance on a varied set of downstream tasks.

3 Results

3.1 NeuralBench-EEG v1.0: Evaluating EEG foundation models on 36 diverse downstream tasks

We summarize per-model performance in Figure 3, where we order the models based on their average per-task rank. While the highest scoring models are foundation models, *i.e.*, REVE (El Ouahidi et al., 2025), LaBraM (Jiang et al., 2024) and LUNA (Döner et al., 2025), some task-specific architectures trained *from scratch* also perform well, *e.g.* CTNet (Zhao et al., 2024), SimpleConvTimeAgg (Défossez et al., 2022) and Deep4Net (Schirrmester et al., 2017), despite significantly lower parameter counts (*e.g.* 150K parameters for CTNet vs. 69.2M for REVE). This is in line with Kastrati et al. (2025) and Wu et al. (2025) which reported similar results. Of note, BENDR (Kostas et al., 2021) and BIOT (Yang et al., 2023) did not perform as well overall despite their pretraining, which might in part be explained by the necessary addition of a channel adapter linear layer to match the expected channel montage seen during pretraining, and, for BIOT, to the use of a linear projection head rather than a large MLP head like in the original publication.

3.2 Future-proofing evaluation through task diversity and difficulty

Detailed performance results for each downstream task and model are presented in Figure 4 and Figure 5. The results suggest that performance over a few common EEG tasks is close to saturation, *e.g.* SSVEP classification, pathology, seizure detection, sleep stage classification and phenotyping tasks, *i.e.*, age regression and sex classification.

In contrast, the cognitive decoding tasks are particularly challenging. These tasks, introduced in NeuralBench (*e.g.* speech, sentence, video, word and image decoding), consist of decoding dense representations of the stimulus or condition from the brain activity. Foundation models particularly stand out on these compared to

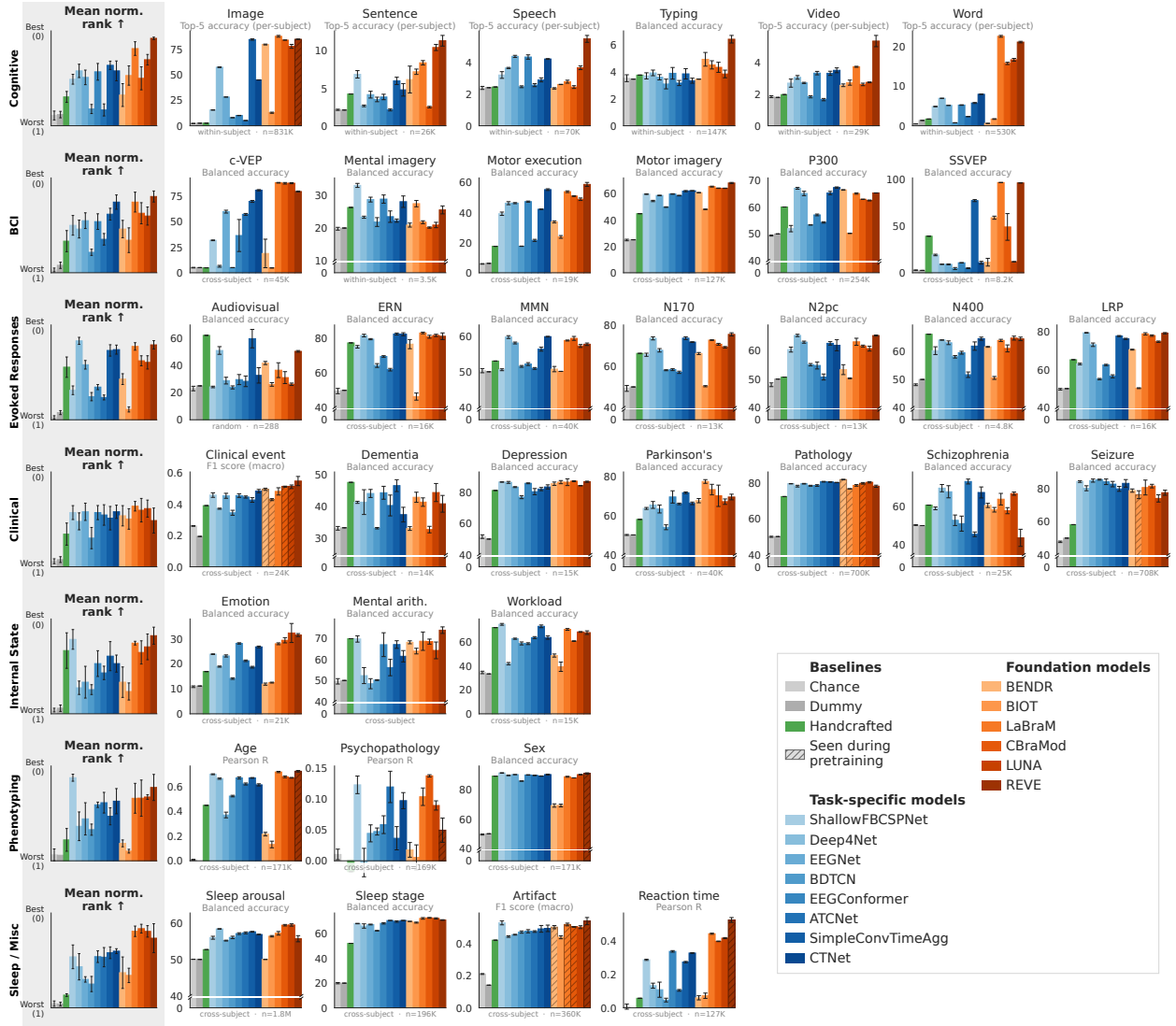


Figure 4 Performance obtained by the different task-specific (blue) and foundation (orange) models on the 36 downstream tasks in NeuralBench-EEG-Core v1.0. Rows correspond to specific downstream task categories; the first subplot of each row summarizes the model ranking over the corresponding task category row. Each bar corresponds to one model, with error bars representing standard error of the mean (SEM) across seeds. Hashed bars indicate models that were pretrained on the downstream dataset, meaning the reported performance could be overinflated since test subjects or examples may have been seen by the model. Dummy-level performance is evaluated for each task by predicting the majority class (classification) or the average target (regression and retrieval) computed on the training set. Handcrafted baselines uses sklearn pipelines which ingest covariance-like features to provide a non-deep learning baseline (see Table S3). Finally, the splitting strategy (cross-subject, within-subject or random) and the total number of examples across splits are listed under each subplot.

task-specific models, *e.g.* REVE leads on the sentence, speech and video decoding tasks.

Models perform significantly worse on NeuralBench’s rarer paradigms, including mental imagery, sleep arousal, and psychopathology decoding, highlighting the difficulty of these specific decoding tasks. Moreover, standard benchmark tasks, *i.e.*, motor imagery, P300 and N2pc classification, become particularly challenging when framed as cross-subject tasks, often yielding performance close to dummy level. Given the substantial margin between current results and performance ceiling, these tasks represent ideal benchmarks for future-proofing the evaluation of the next generation of EEG models.

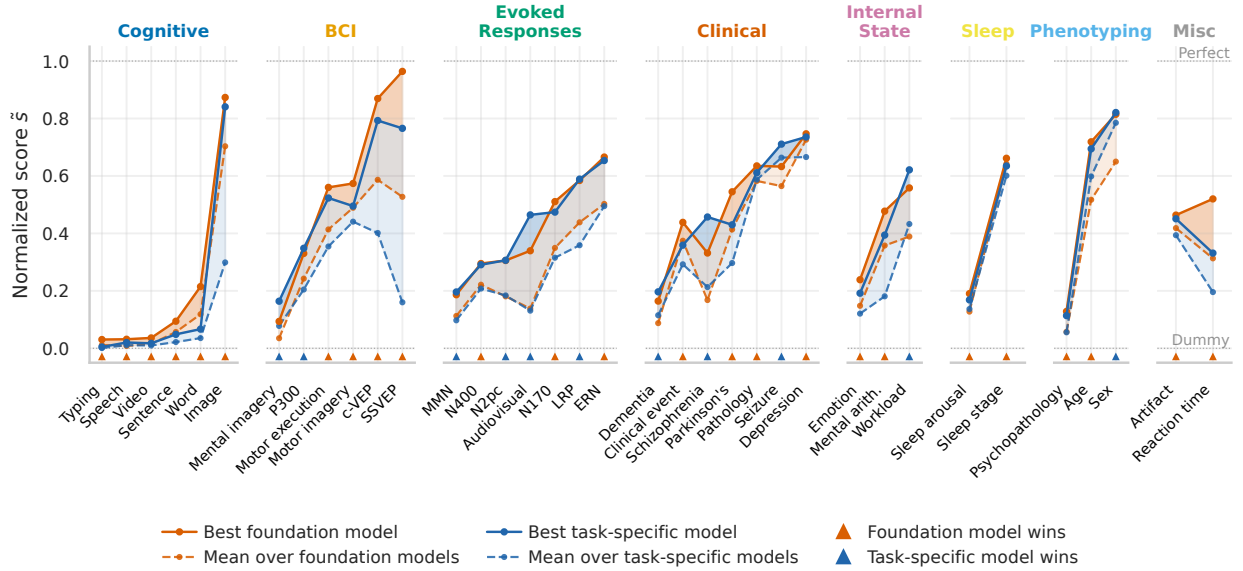


Figure 5 Normalized performance $\tilde{s}$ across downstream tasks, for the best and “average” foundation model (orange) and best and “average” task-specific models (blue). A value of 0.0 corresponds to dummy-level performance, while a value of 1.0 represents perfect (idealized) performance. Horizontal bars on the x-axis indicate which model category yield the highest performance overall for a given task. While foundation models often outperform task-specific models, this is not the case across all tasks, *e.g.* in evoked responses and clinical downstream tasks. Cognitive decoding tasks introduced in **NeuralBench-EEG v1.0** are particularly challenging, which position them well for evaluating future iterations of EEG foundation models.

3.3 From NeuralBench-Core to NeuralBench-Full: Leveraging multi-study tasks for evaluating cross-dataset variability

The results above focused on **NeuralBench-EEG-Core v1.0**, a version of the benchmark where each task is summarized by a single representative “core” dataset. We also introduce **NeuralBench-EEG-Full v1.0**, an extended version of the benchmark in which downstream tasks may include additional datasets on top of their core dataset, which we leverage in two ways.

First, we compare models as above, but this time using dataset-averaged ranks whenever multiple datasets are available for a task, yielding a more robust ordering of models. The “Core” and “Full” rankings are compared in **Figure 6A**. A Kendall’s τ of 0.926 ($p < 0.001$) between the two rankings suggests the “Core” ranking is a good proxy for the “Full” one. However, we notice that a few models swap positions in the ranking, *e.g.* CTNet overtakes the LUNA foundation model as third best in the “Full” ranking. This shows the gap between task-specific and foundation models is narrow enough that expanding dataset coverage is sufficient to change global ranking.

Second, we evaluate within-task variability by inspecting ranking changes in **NeuralBench-EEG-Full v1.0**’s multi-dataset tasks (**Figure 6B**). REVE and LaBraM are the most stable foundation models, with the former experiencing the least rank variability of all models. On the task dimension, we notice strong variability on **motor imagery** and **SSVEP** in particular. For instance, on SSVEP, performance on the core dataset ([Wang et al., 2017](#)) and on [Lee et al. \(2019\)](#) is mostly dominated by foundation models (LaBraM and REVE), task-specific models outperform all other models on [Oikonomou et al. \(2016\)](#) version A, and the handcrafted features-based baseline outperforms all or most other models on [Kalunga et al. \(2016\)](#) and [Oikonomou et al. \(2016\)](#) versions B and C (see Supplementary **Figure S2** for details). This strong variability suggests that multi-study comparisons are necessary to fully characterize the behavior of brain models.

Moving forward, the availability of multiple datasets per task will allow testing generalization and invariance properties of foundation models over varying recording hardware, lab environments and subject populations. It also opens the door to multi-dataset finetuning and evaluation, *e.g.* in which models are finetuned on a

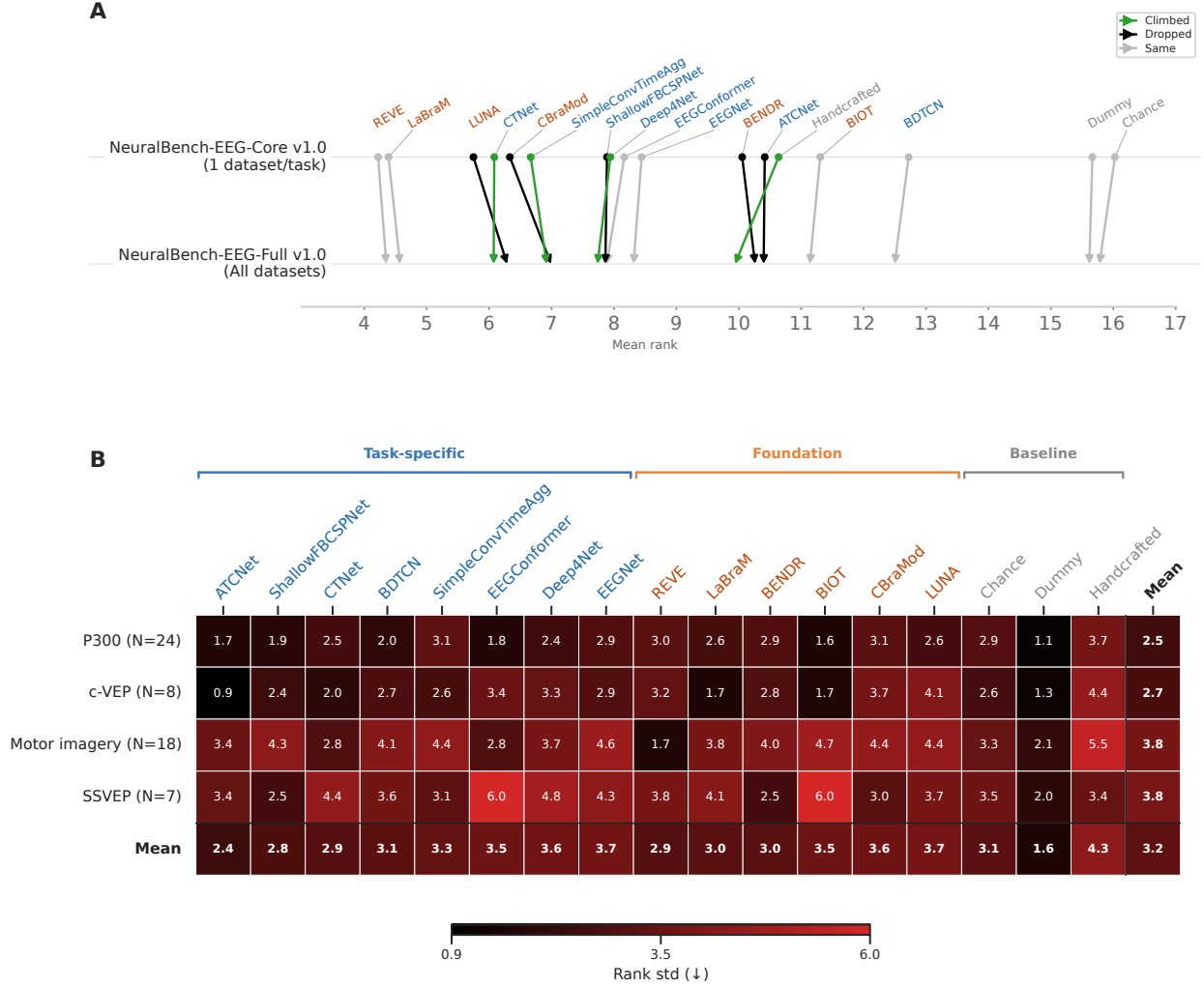


Figure 6 Analysing multi-dataset performance on NeuralBench-EEG-Full v1.0. **A.** Difference in model ranking between the *Core* and *Full* variants of NeuralBench-EEG v1.0. Models whose global rank improve or degrade when using the full version of the benchmark are indicated by green or black lines, respectively. The ranking is not affected much by the addition of multiple datasets per task, which suggests *Core* datasets are overall representative. **B.** Within-task model ranking variability. We report the standard deviation of each model’s ranks across the different datasets of a task (for tasks with 5 datasets or more). Higher values indicate a model’s performance relative to the other models vary substantially, meaning the model is not stable when evaluated across multiple datasets of the same domain. Tasks like motor imagery and SSVEP, and foundation models like CBraMod and LUNA, are the most unstable.

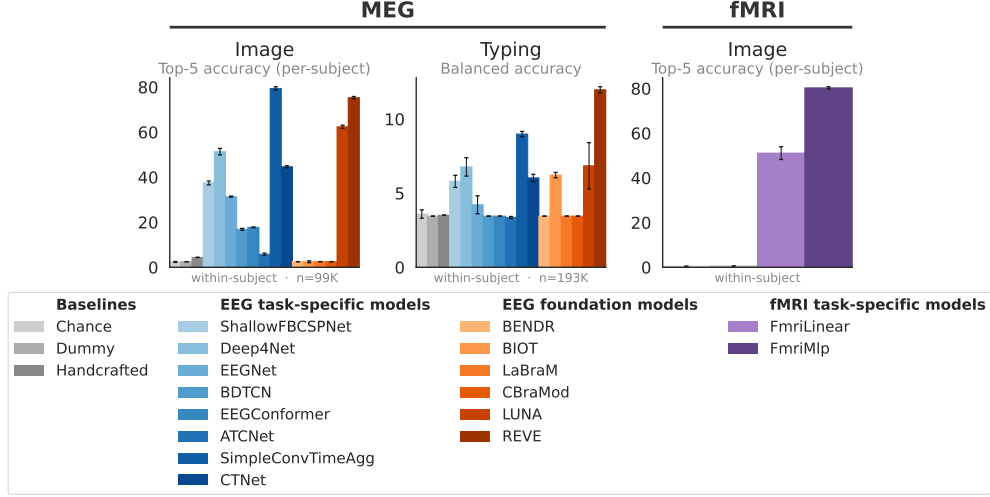


Figure 7 Performance obtained on MEG and fMRI downstream tasks. Each bar corresponds to one model, with error bars representing standard error of the mean (SEM) across seeds. See Figure 4 for more details. This first benchmark release focuses on EEG, however, as a unified framework, NeuralBench is ready to support other brain imaging modalities as well.

subset of task datasets, and evaluated on the remaining held out datasets.

3.4 Extending NeuralBench to more modalities: MEG and fMRI

The focus of this first NeuralBench release is EEG. However, the benchmark infrastructure already provides support for other neuroimaging modalities, which we demonstrate by including results on two MEG and one fMRI downstream tasks (see Figure 7).

For MEG, given the similarities between the two types of data (da Silva, 2013) and the common reuse of EEG architectures on MEG (Banville et al., 2025), we evaluate the same task-specific and foundation models as for EEG. Expectedly, some architectures yield high performance on MEG, such as SimpleConvTimeAgg which was already shown to perform well on image decoding on the same dataset (Benchetrit et al., 2024). More strikingly, some EEG foundation models, despite being pretrained on EEG data exclusively, perform well on MEG, with REVE outperforming all other models on typing decoding. Finally, on fMRI, we present results on a similar image decoding task as in EEG and MEG, but focused on two task-specific architectures that rely on linear and convolutional layer primitives. These results exhibit the flexibility of NeuralBench, which is ready to be extended to other modalities as well, such as iEEG, functional near-infrared spectroscopy (fNIRS), and other neural signal types such as electromyography (EMG).

Overall, in this section, we highlighted NeuralBench’s comprehensive downstream task coverage, its versatility, allowing easy between- and within-task performance comparisons, and its extendability to more neuroimaging devices.

4 Discussion

EEG foundation vs. task-specific models. We introduce NeuralBench, a unifying framework to benchmark NeuroAI models, and release **NeuralBench-EEG v1.0**, a first comprehensive benchmark evaluating existing task-specific and foundation models of EEG signals. Our results already reveal that current EEG foundation models only marginally outperform specialized models. While self-supervised pretraining yields generalizable representations, downstream performance gains remain limited. This modest advantage may stem from the use of finetuning as a downstream strategy: in data-rich scenarios, specialized models likely possess sufficient capacity to close the performance gap with foundation models. Future evaluations should thus also incorporate evaluations in few-shot and low-data regimes (e.g. 5% of training data).

EEG task expansions. NeuralBench-EEG v1.0 significantly expands the diversity of downstream evaluations by introducing and integrating a large set of cognitive, phenotyping, clinical and BCI “tasks”. Across most of these tasks, performance remains modest, which leaves room for improvement in next generations of EEG foundation models. In the future, we recommend maintaining a diverse set of tasks – some easy, others difficult – to validate the ability of models to extract valuable information from noisy brain signals, while ensuring that benchmarking is representative of downstream neuroscience and clinical applications.

Unique optimization strategies. Rather than pursuing model-specific optimization, we utilize a standardized training recipe to evaluate each architecture’s “out-of-the-box” utility. This decision reflects a core requirement for NeuroAI foundation models: they must be easily adaptable and stable under standard configurations to be practically useful for the broader research community.

Unique splitting strategy. We used specific splitting strategies to maintain a manageable computational footprint while ensuring diverse task coverage. Although this choice limits the assessment of performance variability (as compared to a cross-validated approach), it establishes a broad baseline for evaluating foundation models. We invite the community to help incorporate stochastic partitioning in future releases to facilitate deeper statistical validation.

Unique finetuning strategy. NeuralBench currently relies on end-to-end finetuning, which may not capture the full potential of foundation models. Alternative strategies, such as linear probing, parameter-efficient finetuning (Hu et al., 2022; Lester et al., 2021) and varied token aggregation methods may further improve performance and differentiate models from one another. These dimensions should thus be studied in greater detail in future iterations of the benchmark.

Beyond EEG. While this first release focuses on EEG – given its prominence in the brain foundation model literature (Yang et al., 2025) – NeuralBench’s infrastructure based on the NeuralSet ecosystem (King et al., 2026) is highly flexible. We demonstrate this with preliminary MEG and fMRI tasks, paving the way for a truly multimodal benchmark, and strikingly, already showing promising transfer from EEG to MEG. However, extending to new modalities requires significant data curation and introduces device-specific challenges, such as high auto-correlation in fMRI that complicates train/test splits (Poldrack, 2006).

Beyond decoding. The current benchmark is restricted to the evaluation of models which receive brain activity as input, and predict a target label or embedding, *i.e.*, *decoding* models. However, *encoding* models (Naselaris et al., 2011), which predict brain activity from stimuli, are also actively being developed. Existing benchmarking efforts in the neuroAI community, including machine learning competitions, are focused on encoding, *e.g.* the Algonauts project (Cichy et al., 2021; Gifford et al., 2023, 2024) on image and video encoding in human fMRI, the Sensorium competition (Willeke et al., 2022; Turishcheva et al., 2024) on image and video encoding in mouse visual cortex calcium imaging, and the Brain-Score platform (Schrimpf et al., 2018, 2020) on visual and language encoding in primate electrode recordings and human fMRI. Integrating encoding, forecasting, and denoising tasks presents unique challenges, *e.g.* standardizing stimuli representations and evaluation metrics, but can be readily accommodated within the NeuralBench framework.

Open call for community contribution. Building a universal benchmark for brain activity is a vast undertaking, unlikely to be achieved by a single team. We therefore open-source the NeuralBench codebase¹⁶ and invite the community to contribute new datasets, tasks, models, and neuroimaging modalities.

Towards a unified modeling of brain activity. Ultimately, brain foundation models have the potential to revolutionize brain-computer interfacing, neurology, and neuroscience. Moving toward the integrated processing of multimodal brain data requires rigorous, reproducible, and comprehensive evaluation. We hope NeuralBench serves as a foundational tool to seed and support this community-wide effort.

¹⁶<https://github.com/facebookresearch/neuroai/tree/main/neuralbench-repo>

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Appendix

A Additional details on downstream datasets and tasks

We provide the references for all datasets used in NeuralBench-EEG v1.0 (and the MEG and fMRI extensions) in Table S1. Detailed information about the *Core* datasets for each task can be found in Table 1. We do not repeat references when multiple datasets come from the same paper.

Of note, these datasets sometimes contain stimuli presented to the participants (*e.g.* images, videos, text, etc.). We only make use of stimuli in the cognitive decoding tasks (Gifford et al., 2022b; Hollenstein et al., 2018; Brennan and Hale, 2019; Liu et al., 2024b; Nieuwland et al., 2018), as these tasks require the alignment of brain data to a dense representation of the stimuli. Apart from these cases, we discard stimulus information and only keep targets (*e.g.* *classes* for classification tasks).

Detailed and up-to-date description of the downstream tasks can be found in the benchmark documentation at <https://github.com/facebookresearch/neuroai/tree/main/neuralbench-repo>.

B Task-specific architectures and foundation models for EEG downstream tasks

We list the task-specific deep-learning architectures, handcrafted features-based sklearn baselines, and foundation models used on the EEG downstream tasks in Tables S2, S3 and S4.

C Detailed splitting strategy

We split each dataset into a single training, validation and testing partition using a fixed seed. The way we split depends on the datasets. When datasets explicitly provide a development and evaluation split, we use it as is. This is the case for **clinical event classification** on TUEV (Harati et al., 2015), **pathology detection** on TUAB (Lopez et al., 2015), **image decoding** on THINGS-EEG2 (Gifford et al., 2022b) and **motor imagery classification** on the BEETL AI Challenge datasets (Wei et al., 2022).

For cognitive decoding experiments, we focus on “leave-concept-out” evaluation, *i.e.*, all subjects are seen in both training and testing splits, but a common subset of the sessions or concepts are held out for testing. In many cases, this setting is required to obtain performance significantly above chance-level, as cross-subject generalization is particularly challenging on these types of tasks. For **speech decoding** on Brennan and Hale (2019), we keep the last 4 runs of each recording as test set. For **sentence decoding** on ZuCo v1 (Hollenstein et al., 2018), we randomly hold out 20% of the sentences read by the subjects to use as test set. For **word decoding** on Nieuwland et al. (2018), we randomly hold out 10% of the sentences read by the subjects to use as test set. For **video decoding** on SEED-DV (Liu et al., 2024b), we keep a random 20% of the video concepts (*e.g.* land animal, human, natural scene, etc.) to use as test set. For **typing decoding** on (Lévy et al., 2025), sentences are clustered using their similarity in sentence embedding space (TF-IDF), and 20% of similar sentences are held out for testing.

Most remaining tasks/datasets use a cross-subject split. For **age**, **psychopathology**, **sex** and **reaction time prediction**, all on the HBN-EEG dataset (Shirazi et al., 2024), we leave out all recordings from release 5 (out of 11 releases) for testing, which corresponds to ~ 329 held-out subjects. For other tasks, we hold out a random 20% of the subjects for testing. Of note, this cross-subject strategy is particularly challenging for certain tasks as inter-subject variability renders generalization difficult, *e.g.* in BCI (Saha and Baumert, 2020). Moreover, on clinical diagnosis tasks where subjects receive a single diagnosis (**dementia**, **depression**, **Parkinson’s disease** and **schizophrenia**), we use stratified splits based on the diagnosis label.

A few datasets instead use a within-subject split by default to accommodate low (<10) subject counts, by keeping the last (few) sessions or runs for testing (Zhou et al., 2016; Chavarriaga and Millan, 2010; Hoffmann et al., 2008; Kueper et al., 2024).

Finally, we use fully random splits, *i.e.*, at the example level, for three low-sample datasets: **audiovisual stimulus** classification on the MNE Sample dataset (Gramfort et al., 2013) and **motor imagery** classification on the datasets from Dornhege et al. (2004); Barachant (2012).

Validation sets (used for early stopping) are sampled from the remaining data, using the same or a similar splitting strategy as used for the test set, targeting another 20% of the overall dataset size. We train each model on each task three times, using three different random seeds for the initialization of the (non-pretrained) model parameters.

D Performance metrics

We pick a single representative metric per task type to report in this document. For binary and multiclass classification, we use the balanced accuracy:

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c} \quad (\text{S1})$$

where C is the number of classes, TP_c is the number of true positives for class c , and FN_c is the number of false negatives for class c .

For multilabel classification, we use the macro F1-score:

$$\text{Macro F1} = \frac{1}{C} \sum_{c=1}^C F1_c = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c} \quad (\text{S2})$$

where $P_c = \frac{TP_c}{TP_c + FP_c}$ is the precision and $R_c = \frac{TP_c}{TP_c + FN_c}$ is the recall for class c .

For univariate regression, we use the Pearson correlation:

$$r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (\text{S3})$$

where y_i are the true values, $\hat{y}_i$ are the predicted values, $\bar{y}$ and $\bar{\hat{y}}$ are their respective means, and n is the number of samples.

Finally, for retrieval, we use the top-5 accuracy:

$$\text{Top-}k \text{ Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[y_i \in \hat{Y}_i^{(k)} \right] \quad (\text{S4})$$

with $k = 5$, where N is the number of queries, y_i is the true label (or relevant item) for query i , $\hat{Y}_i^{(k)}$ is the set of the top- k retrieved candidates for query i , and $\mathbb{1}[\cdot]$ is the indicator function.

More metrics are computed in NeuralBench, *e.g.* root mean-squared error (RMSE) for regression and median ranks for retrieval, but are not reported here for brevity.

E Additional results on NeuralBench-EEG-Core v1.0

Figure S1 shows max-normalized scores over core datasets.

F Additional results on NeuralBench-EEG-Full v1.0

Figure S2 shows performance on each multi-dataset downstream task included in NeuralBench-EEG-Full v1.0.

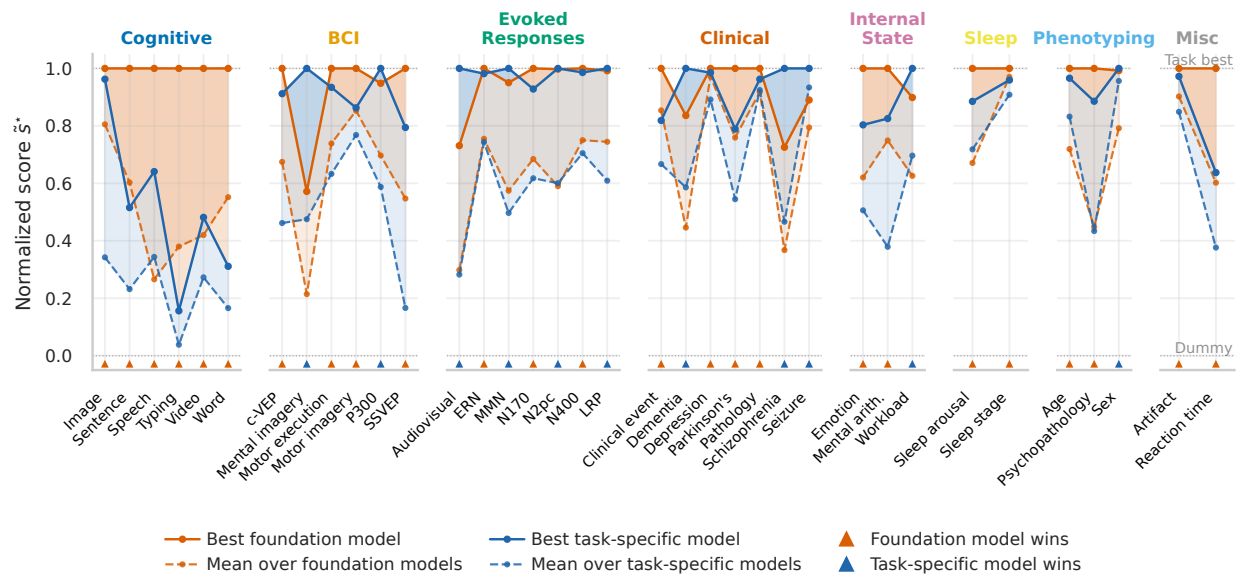


Figure S1 Max-normalized performance $\tilde{s}^*$ across downstream tasks, for the best and “average” models. As in Figure 5, but scores are normalized to be between “dummy” performance and per-task best model performance, highlighting variability as compared to the highest performing model.

Table S1 Dataset references for all NeuralBench-EEG v1.0 tasks, grouped by category. References for the *primary* dataset are listed first; additional dataset references follow.

Category	Task	Dataset references
Cognitive	Image	Gifford et al. (2022a), Hebart et al. (2023), Allen et al. (2022)
	Sentence	Hollenstein et al. (2018)
	Speech	Brennan and Hale (2019)
	Typing	Lévy et al. (2025); Zhang et al. (2025)
	Video	Liu et al. (2024b)
	Word	Nieuwland et al. (2018)
BCI	c-VEP	Cabrera Castillos et al. (2023), Martínez-Cagigal (2023), Martínez-Cagigal (2024), Thielen et al. (2015), Thielen et al. (2021)
	Mental imagery	Scherer et al. (2015)
	Motor execution	Srisrisawang and Müller-Putz (2024), Schirrmeister et al. (2017), Ofner et al. (2017)
	Motor imagery	Stieger et al. (2021), Schalk et al. (2004), Barachant (2012), Cho et al. (2017), Dornhege et al. (2004), Dreyer et al. (2023), Faller et al. (2012), Grosse-Wentrup et al. (2009), Lee et al. (2019), Leeb et al. (2007), Liu et al. (2024a), Schwarz et al. (2020), Shin et al. (2017), Scherer et al. (2012), Tangermann et al. (2012), Wei et al. (2022), Yi et al. (2014), Zhou et al. (2016)
	P300	Schreuder et al. (2010), Kappenman et al. (2021), Acqualagna and Blankertz (2013), Aricò et al. (2014), Cattani et al. (2019), Vaineau et al. (2018), Guger et al. (2009), Haufe et al. (2011), Hoffmann et al. (2008), Hübner et al. (2017), Hübner (2016), Kojima and Kanoh (2024), Kojima (2024), Korczowski et al. (2019b), Korczowski et al. (2019c), Korczowski et al. (2019a), Lee et al. (2019), Riccio et al. (2013), Romani et al. (2025), Schaeff et al. (2012), Höhne (2011), Sosulski and Tangermann (2019), Treder et al. (2011), Treder et al. (2014), Van Veen et al. (2019)
	SSVEP	Wang et al. (2017), Lee et al. (2019), Kalunga et al. (2016), Nakanishi et al. (2015), Oikonomou et al. (2016)
Evoked Responses	Audiovisual stimulus	Gramfort et al. (2013)
	Error-related negativity	Kappenman et al. (2021), Kueper et al. (2024), Chavarriaga and Millan (2010)
	MMN	Kappenman et al. (2021)
	N170	Kappenman et al. (2021)
	N2pc	Kappenman et al. (2021), Reichert et al. (2020)
	N400	Kappenman et al. (2021)
Clinical	Lateralized readiness potential	Kappenman et al. (2021)
	Clinical event detection	Harati et al. (2015)
	Dementia diagnosis	Miltiadous et al. (2023)
	Depression diagnosis	Mumtaz et al. (2018)
	Parkinson's diagnosis	Singh et al. (2021)
	Pathology	Lopez de Diego (2017)
Internal State	Schizophrenia diagnosis	Albrecht et al. (2019)
	Seizure detection	Dan and Shoeb (2023)
	Emotion recognition	Chen et al. (2023)
	Mental arithmetic	Zyma et al. (2019), Shin et al. (2017)
	Mental workload	Hinss et al. (2022), Hinss et al. (2023), Jao et al. (2023)
Sleep	Sleep arousal	Ghassemi et al. (2018)
	Sleep stage	Kemp et al. (2000)
Phenotyping	Age	Shirazi et al. (2024)
	Psychopathology	Shirazi et al. (2024)
	Sex	Shirazi et al. (2024)
Misc	Artifact detection	Hamid et al. (2020)
	Reaction time	Shirazi et al. (2024)

Table S2 Task-specific EEG models evaluated in **NeuralBench-EEG v1.0**. Parameter counts are backbone-only (excluding the task-specific output head) and are approximate as they depend on the number of input channels (we use a representative value of $C=22$) here.

Model	# Params	Reference
ShallowFBCSPNet	36K	Schirrmester et al. (2017)
Deep4Net	146K	Schirrmester et al. (2017)
EEGNet	1.5K	Lawhern et al. (2018)
BDTCN	27K	Gemein et al. (2020)
EEGConformer	277K	Song et al. (2022)
ATCNet	29K	Altaheri et al. (2022)
SimpleConvTimeAgg	4.2M	El Ouahidi et al. (2023)
CTNet	150K	Zhao et al. (2024)

Table S3 Handcrafted features-based sklearn baselines evaluated in **NeuralBench-EEG v1.0**. These pipelines are fit once on the concatenation of the training and validation sets and evaluated with the exact same preprocessing and splits as the deep learning models. A single pipeline is used per task type; see *Tasks* column.

Pipeline	Description	Tasks	Reference
XdawnTsLR	xDAWN covariances $\rightarrow$ shrinkage $\rightarrow$ tangent space $\rightarrow$ standard scaling $\rightarrow$ logistic regression (CV-tuned C)	Classification: <i>Evoked response</i> and <i>P300</i>	Rivet et al. (2009) , Barachant and Congedo (2014)
CovTsLR	sample covariances $\rightarrow$ shrinkage $\rightarrow$ tangent space $\rightarrow$ standard scaling $\rightarrow$ logistic regression (CV-tuned C ; one-vs-rest for multilabel)	Classification: <i>BCI</i> (except <i>P300</i> and <i>SSVEP</i>), <i>Internal State</i> , <i>Clinical</i> , <i>Sleep</i> , <i>sex</i> , <i>artifact</i>	Barachant et al. (2011)
CoSpectraLogLR	Welch co-spectra $\rightarrow$ upper-triangular flatten with $\log(1+x)$ on the diagonal (per-channel PSD) $\rightarrow$ standard scaling $\rightarrow$ logistic regression (CV-tuned C)	Classification: <i>SSVEP</i>	Kalunga et al. (2016)
CovTsRidge	sample covariances $\rightarrow$ shrinkage $\rightarrow$ tangent space $\rightarrow$ standard scaling $\rightarrow$ ridge regression (CV-tuned α)	Regression: <i>age</i> , <i>psychopathology</i> , <i>reaction time</i> ; Retrieval (treated as multivariate regression here): <i>Cognitive</i>	Barachant et al. (2011) , Hoerl and Kennard (1970)

Table S4 EEG foundation models evaluated in **NeuralBench-EEG v1.0**. Parameter counts are backbone-only (excluding the task-specific linear probe). Pretraining data quantities are reported as published; TUEG refers to the Temple University Hospital EEG Corpus.

Model	Variant	# Params	Pretraining	Data	Reference
BENDR	—	157.1M	Wav2vec-like contrastive	TUEG, ~ 1.5 K h	Kostas et al. (2021)
BIOT	6-datasets-18chs	3.2M	Contrastive + supervised	6 datasets (CHB-MIT, TUAB, TUEV, +3 private)	Yang et al. (2023)
LaBraM	Base	5.8M	VQ-VAE masked prediction	16 datasets, ~ 2.5 K h	Jiang et al. (2024)
CBraMod	—	4.9M	Masked patch reconstruction	TUEG, ~ 27 K h	Wang et al. (2025b)
LUNA	Large	40.4M	Masked patch reconstruction	TUEG + Siena	Döner et al. (2025)
REVE	Base	69.2M	Masked patch reconstruction	92 datasets, ~ 60 K h	El Ouahidi et al. (2025)

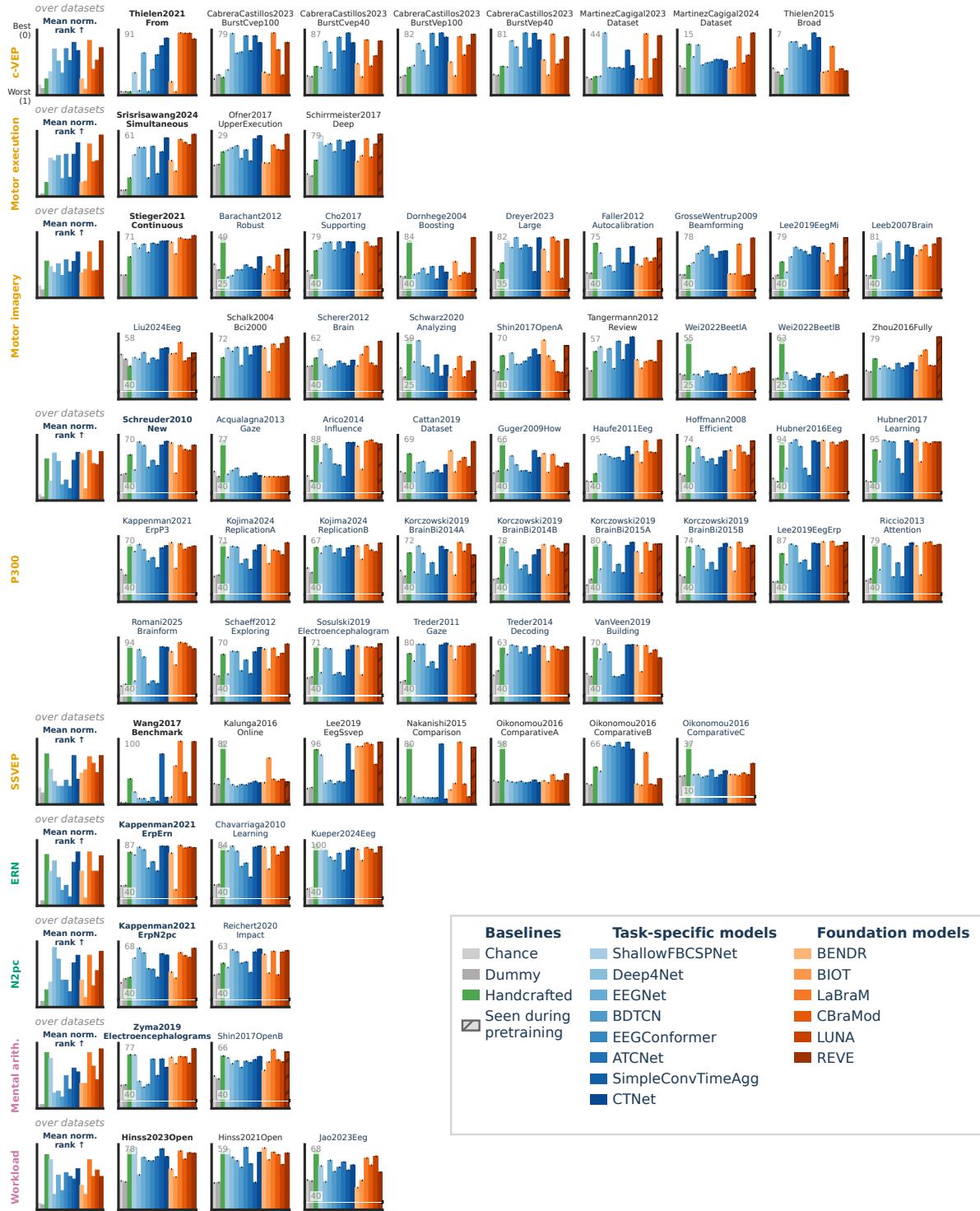


Figure S2 Performance obtained by the different task-specific (blue) and foundation (orange) models on the multi-dataset NeuralBench-EEG-Full v1.0 downstream tasks (see Figure 4’s caption for details).